Check for updates

Empirical Research

Hammill Institute on Disabilities

The Journal of Special Education
1–12
© Hammill Institute on Disabilities 2021
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/00224669211032556
journalofspecialeducation.sagepub.com
SAGE

Increasing Social Interactions of Preschool Children With Autism Through Cooperative Outdoor Play

Sonja M. T. Ziegler, MS, Dipl.-Psych, BCBA[1,2,3] and Michael J. Morrier, PhD, BCBA-D[1]

Abstract

Unstructured activity periods are central components of early childhood education programs. Deficits in social communication and social interaction presented by children with autism spectrum disorder (ASD) become apparent in these settings during unstructured activities, such as outdoor recess. Thirty-two preschool-aged children with and without ASD participated in the *Buddy Game*, a structured, cooperative play curriculum that focused on natural peer–peer modeling and imitation. The *Buddy Game* used familiar songs to promote peer proximity and discrete social bids between peers during outside time. A multiple baseline design across three classrooms was used to study curriculum effects on discrete social bids of children with ASD. Results demonstrated increased social bids from children with ASD to their peers with neurotypical development (NTD) across classrooms. In one classroom, effects maintained after intervention stopped and generalized to other times of the day. Implications for integrated educational settings and early childhood curricula are discussed.

Keywords

autism spectrum disorder, preschool-aged children, outdoor recess curriculum, peer social interactions, neurotypical developing peers

Children diagnosed with autism spectrum disorder (ASD) present deficits in key developmental areas of social communication and interaction, and unusually restricted, repetitive behaviors, and interests (American Psychiatric Association [APA], 2013). Current prevalence estimates of ASD indicate a rate of 1 in 64 for 4-year-old children (Shaw et al., 2020). Thus, early child care settings have an increased chance of having a child with ASD enrolled. In such settings, the developmental deficits of children with ASD affect their ability to interact with teachers and peers across various contexts, including the classroom, transition areas, meals, and recess.

Early childhood settings are required to offer daily outdoor recess times (Georgia Department of Early Care and Learning, 2016); however, the curriculum during this time is commonly unstructured, with staff numbers reduced to offer adult break times, lending less adult contact opportunities for children (Kasari et al., 2012). This lack of structured activity time, which could be used for focused treatment, or to provide a clear opportunity for targeted social interaction, frequently exacerbates the social deficits of children with ASD (McNamara et al., 2018).

During the early childhood years, play is the natural context in which children with neurotypical development (NTD) learn and expand their language and social skills. Research has demonstrated that social communication interventions for children with ASD can be effectively embedded into play activities (Boyd et al., 2008; McGee & Daly, 1999; McGee et al., 2001; Morrier et al., 2009; Stahmer et al., 2011; Strain & Bovey, 2011). By teaching children with NTD how to play with their peers with ASD, increases in social bids from children with ASD and NTD can be achieved (Harper et al., 2008; Kohler et al., 2007; Laushey & Heflin, 2000; McGee et al., 1992; Owen-DeSchryver et al., 2008). Furthermore, NTD children who are taught to respond to social bids from their peers with ASD results in increased frequency and topography of bids from those peers (Lang et al., 2011). McEvoy et al. (1988) demonstrated that embedding interactive songs and games

[1]Emory University, Atlanta, GA, USA
[2]Georgia State University, Atlanta, GA, USA
[3]Børne og Ungdomspsykiatri Odense, Denmark

Corresponding Author:
Sonja M. T. Ziegler, Psychiatrien i Region Syddanmark, Børne og Ungdomspsykiatri Odense, J.B. Winsløws Vej 28 indgang 236, Odense 5000, Denmark.
E-mail: smt.ziegler@gmail.com.

that have affection actions into typical early childhood circle time activities increased social bids of children with ASD. Other studies demonstrated the benefits of pivotal response training at recess (e.g., turn taking, appropriate commenting), during which children with ASD increased their social bids to peers with NTD (Harper et al., 2008; Koegel et al., 2006). Children with ASD have also shown language and social gains when intervention was provided during games that included child choice of what to do next (Carter, 2001; Miller & Almon, 2009).

However, previous research on increasing social interactions between children with ASD and their peers with NTD has focused on elementary-aged students, and on only two to three target children per study (Boyd et al., 2008; Harper et al., 2008; McEvoy et al., 1988; Owen-DeSchryver et al., 2008). Peer-to-peer interaction research conducted in inclusive preschool (PS) settings has previously included children aged 3 years 7 months and older, but not younger (McGee et al., 1992). Several studies that included paired ASD/NTD dyads highlighted procedural weaknesses. Owen-Deschryver et al. (2008) described selecting new peers with NTD for a participant with ASD, "after it was determined that the training with these two peers was not succeeding in increasing peer interactions" (p. 18). Drops in procedural fidelity between PS peer dyads were specifically described by McGee et al. (1992): ". . . the peer tutors (with NTD) often seemed to forget their 'teaching responsibilities' as they became engaged in play with the target peers and mutually interesting toys" (p. 124). In addition, some procedures implemented by McGee et al. (1992) were both developmentally inappropriate and difficult for peers to replicate in their natural social interactions. Issues raised included directive prompts for language, followed by arbitrary responses by peers (e.g., "say car," "good saying car"), which explicitly placed the peer in the role of teacher and the child with ASD in the role of student. Although comprehensive early intervention programs for PS-aged children with ASD have provided naturally occurring social interaction opportunities (McGee et al., 1999, 2001; Strain & Bovey, 2011), recess is often an unstructured free-play period, rather than a structured period in which children are supported to experience positive social interactions (Kasari et al., 2012).

The *Buddy Game*

To overcome the social communication and social interaction impairments associated with ASD during outdoor recess, the first author designed a novel structured outdoor play curriculum, the *Buddy Game*, specifically targeting impairments in social interactions between children with ASD and their NTD peers. The *Buddy Game* was designed to use a structured, cooperative play activity that used naturally occurring opportunities for peer modeling and imitation (McEvoy et al., 1988; Schreibman et al., 2015) to promote positive social initiations (i.e., social bids) from children with ASD *to* their NTD peers (Lang et al, 2011). By pairing children's preferred songs with movement activities (McGee et al., 1992; Morrier et al., 2009), children were motivated to learn how to imitate NTD models and initiate social interactions to their NTP peers on the playground (Carter, 2001; Miller & Almon, 2009).

Outdoor recess was targeted for the *Buddy Game*, due to the American Academy of Pediatrics' recent statement that recess promotes social and emotional learning through opportunities for children to engage in peer interactions in which they practice essential social skills (Council on School Health, 2013). When this type of activity is supervised by adults, it can extend classroom teaching (Harper et al., 2008). Anderson et al. (2004) and McNamara et al. (2018) indicated that in educational settings, children with ASD's social deficits can be clearly observed on the playground. Peeters (1997) and Veale (1998) reported that children with ASD spend most of recess time isolated, socially disconnected, and engaging in self-stimulatory behaviors whose consequences may be more reinforcing, and in turn become more motivating, than functional play with others. Ingram et al. (2007) observed that in an integrated recess setting, many children with ASD were socially isolated, did not engage in social play, and few initiated social interactions with peers.

Previous research on the *Buddy Game* examined the effects of age (in 6-month intervals) and diagnosis (ASD vs. NTD) as they related to increases in social interactions for 35 children (Morrier & Ziegler, 2018). Results of this analysis indicated that increases in social initiations were more prevalent in preschoolers with ASD than in their NTD peers, regardless of severity level and age of the child with ASD.

This study aimed to extend the research on interventions to improve social interactions between PS-aged children with ASD and their NTD peers by also including children under 36 months of age. We examined the effect of the *Buddy Game* on discrete social bids of children with ASD toward their NTD peers in three different PS classrooms. Specifically, we evaluated: (a) if the target behaviors increased in all three classrooms following the introduction of the *Buddy Game*, (b) if increases in the targeted behaviors generalized to a different time and setting in all three classrooms, and (c) if increases in the targeted behaviors maintained 1-week post intervention in all three classrooms.

Method

All procedures performed were conducted in accordance with ethical standards of and were reviewed and approved by the Institutional Review Boards of Georgia State University and Emory University. Informed consent was

obtained from all individual adult participants and from a parent/legal guardian for all children in this study.

Experimental Design

The effects of the *Buddy Game* were evaluated using a multiple baseline across classrooms design (Baer et al., 1968; Gast & Ledford, 2014). Three inclusive classrooms of children with and without ASD participated, and the *Buddy Game* was introduced to each sequentially once the previous classroom's baseline data were stable. Each classroom subsequently moved from intervention into the maintenance phase when participants with ASD demonstrated a frequency of discrete social bids greater than or equal to the average number of discrete social bids from NTD children to peers with NTD in baseline for three consecutive days.

Participants and Settings

Participating classrooms. The study was conducted in three of four PS classrooms at the Early Emory Center for Child Development and Enrichment. Each classroom enrolled children with ASD and NTD at a ratio of 1:2. Children were enrolled based on their age on September 1 of that year: early preschool (EPS) enrolled 16 children aged 2 years 0 months to 3 years 0 months; PS enrolled 18 children aged 3 years 0 months to 4 years 0 months; and pre-kindergarten (PK) enrolled 18 children aged 4 years 0 months to 6 years 0 months. Children with ASD were enrolled based on meeting published cut-off scores for autism or autism spectrum on the *Autism Diagnostic Observation Schedule, 2nd edition* (ADOS-2; Lord, Luyster, et al., 2012; Lord, Rutter, et al., 2012) and meeting or exceeding domain cut-off on the *Autism Diagnostic Interview–Revised* (ADI-*R*; Rutter et al., 2003). Thus, children were enrolled regardless of functioning level and each classroom enrolled children across the entire range of ASD.

Teacher-to-child ratio increased from 1:3 in EPS to 1:4 in PS to 1:6 in PK to systematically fade out adult support and increase peer–peer interactions. All classroom curricula used the framework of Naturalistic Developmental Behavioral Interventions (NDBI; Bruinsma et al., 2020; Schreibman et al., 2015). In sum, teachers used applied behavior analysis procedures within natural early childhood PS activities to teach functional developmentally appropriate skills within the activities and routines in which children will need to use those skills.

Participating children. Thirty-two children participated, nine with a diagnosis of ASD and 23 with NTD. Children ranged in age from 2 years 6 months to 6 years 6 months. Due to excessive absences from the program, four NTD children were dropped from the study, reflecting an attrition rate of 14.81%. Prior to initial program entry, diagnoses of children with ASD were confirmed using the ADOS-2 and the ADI-R. Prior to participating, the adaptive behavior of children with ASD was assessed using the *Vineland Adaptive Behavior Scales, 3rd edition* (VABS-3; Sparrow et al., 2016). Characteristics of all child participants can be found in Table 1.

Adult interventionists. Fifteen adults participated in this study: eight lead teachers and seven assistant teachers. Two teachers had associate's degrees, nine had bachelor's degrees, and three had master's degrees. One assistant teacher withdrew from the study due to termination of her employment; reflecting 14.29% attrition. All adults (100%) identified as female, with a mean age of 24.5 years (SD = 2.36), and mean employment time at the center was 21.7 months (SD = 11.61). Thirty percent identified as Asian/Asian American, 47% as Black/African American, 20% as White, 13% as Hispanic/Latin*x*, and 7% as 2 or more races.

Adult interventionist training. Upon consent to participate, the first author met with the participating classroom teachers to explain the intervention and share the procedural fidelity checklists as visual learning supports. Participants then role-played the intervention in the group, with a focus on each role, both inside the classroom and outside on the playground. Training continued until all stated that they felt confident with the procedure.

Settings and materials

Intervention setting. The *Buddy Game* intervention took place on the outdoor playground during each classroom's regularly scheduled, individual, 45-min daily recess period, between minutes 15 and 30 of this period. This time window was chosen to allow children some unstructured playtime while teachers prepared for the *Buddy Game* activity. Using the middle 15-min segment of recess, the class was able to provide ample time to transition back inside the classroom at the end of the recess period.

All classrooms opened directly onto the playground, measuring 139-ft by 34-ft, containing swings and a plastic playhouse; sandbox with toys; two secured bouncing ride-on toys; a wooden play structure with two stairwells and slides; and a multi-seat Spring-a-Bout™. There was also an open area referred to as "the field." The intervention took place on this 37-ft by 34-ft field portion of the playground where no other playground materials were located.

Generalization setting. To measure generalization to a different time of day, in a different setting, daily data were collected during one of each classroom's indoor free play/center time sessions, which occurred after their outdoor recess period. In EPS and PS, one half of the room was considered the "free play" area, which was separated from the "kitchen" area with a glass wall and gate. The free play area

Table 1. Demographic Characteristics of Child Participants.

	Participants with ASD (n = 9)		Participants with NTD (n = 23)	
Variable	*n* (%) *M* (*SD*)	Range	*n* (%) *M* (*SD*)	Range
Gender (%)				
Female	3 (33)		7 (30)	
Male	6 (67)		16 (70)	
Age at time of study (mon; *SD*)	58.67 (10.91)	29–76	39.64 (12.77)	24–66
Time in Program (mon; *SD*)	26.11 (12.58)	10–46	13.58 (11.67)	0–48
Race/ethnicity (%)				
Asian			4 (17)	
Black	4 (44)			
Latin*x*	1 (11)		1 (4)	
White	3 (33)		15 (65)	
NR	1 (11)		3 (13)	
Classroom (%)				
EPS	3 (33)		10 (43)	
PS	3 (33)		5 (22)	
PK	3 (33)		8 (36)	
ADOS-2 (entry; *SD*)				
SA	16.00 (3.30)	11–20		
RRB	5.50 (2.39)	2–8		
Overall	21.50 (4.66)	15–28		
ADI-R (entry)				
Soc	17.57 (4.04)	12–22		
NV Com	11.80 (1.92)	9–14		
Ver Com	19.00 (2.83)	17–21		
RRB	3.71 (1.98)	1–7		
Dev	3.57 (.79)	3–5		
VABS-3 (at time of study)				
Com SS	75.63 (14.94)	47–96		
DLS SS	73.13 (12.29)	57–87		
Soc SS	71.75 (15.23)	48–96		
Mot SS	74.75 (9.21)	61–89		
ABC SS	72.63 (8.82)	60–82		

Note. N = 32. ASD = autism spectrum disorder; NTD = neurotypical development; mon = months; NR = not reported; EPS = early preschool classroom; PS = preschool; PK = pre-kindergarten; ADOS-2 = Autism Diagnostic Observation Schedule, 2nd edition (Lord, Luyster, et al., 2012; Lord, Rutter, et al., 2012); SA = social affect raw score; RRB = restricted, repetitive behavior raw score; overall = overall (Sa + Rrb) raw score; ADI-R = Autism Diagnostic Interview–Revised (Rutter et al., 2003); Soc = social raw score; NV Com = nonverbal communication raw score; Ver Com = verbal communication raw score; Dev = developmental delays before 36 months; VABS-3 = *Vineland Adaptive Behavior Scales, 3rd edition* (Sparrow et al., 2016); Com SS = communication domain standard score; DLS SS = daily living skills domain standard score; Soc SS = social domain standard score; Mot SS = motor domain standard score; ABC SS = adaptive behavior composite standard score.

contained cubbies for children's belongings, and a variety of age appropriate toys, books, and furniture. All children were in this area during generalization, while non-participating children engaged in small group classroom activities on the kitchen side of the room. In PK, seven play interest areas were dispersed throughout the entire classroom in centers rather than all in one free play area. PK centers consisted of manipulatives, kitchen dramatic play, science and sensory, a weekly rotating dramatic play, writing/art, blocks, and books/library. Rules were in place defining how many children could play in each center at one time (2–4), as well as how and when children could move from center to center.

Data collection materials. Data were collected on the behavior of both children with ASD and NTD. Each child was observed individually on a rotating schedule. During the individual observation period, the child was referred to as the target participant. Observers collected frequency data on targeted behaviors directed to the target participant (i.e.,

receipt of discrete social bid) and directed from the target participant (i.e., initiation of discrete social bids) during the child's assigned observation interval using a data collection sheet that included a randomly ordered list of all target participants to be observed, with columns for frequency notation of the dependent variable next to their name.

Intervention materials. Materials included participant nametags, a black poster board with Velcro™ dots, titled the "Buddy Board," and song cards. Participant nametags, made with the child's name and a corresponding animal printed on it, were familiar as they were used in the Center daily, and used to identify daily buddy dyads on the Buddy Board. Song cards had the title of a song printed on it, along with a related picture. Songs were chosen specifically by the first author for their content of gestures and movements, and their familiarity to the child participants. Songs included, "Head, Shoulders, Knees and Toes," "If You're Happy and You Know It," "Baby Shark," "The Banana Song," "The Itsy-Bitsy Spider," "Wheels on the Bus," and "Baby Bumble Bee."

Procedures

During each study phase, observational data were collected in vivo during the second 15-min period of each classroom's 45-min recess by one or two observers. When observing a period simultaneously, observers were each given the same data sheet from which interobserver agreement (IOA) could also be calculated; each observer collected data a minimum of 3 ft apart from each other. Each target participant was observed individually for one 120-s interval during the 15-min observational period, in the order indicated by the data sheet. Data collection procedures were identical for all target participants. Each observer listened to a set of prerecorded prompts on individual Olympus Digital Voice Records (model # VN-100) using individual ear buds. Each recording started with "Ready," followed by 20-s of silence for observers to find the target participant to be observed and orient themselves as needed around the observational field. When observing simultaneously, observers ensured the recordings synchrony by counting "three, two, one," aloud together and simultaneously pressing their play buttons. Observers were prompted to begin an observation interval with "Observe," and to terminate their observation with "Record." Frequency data were collected using a recording system consisting of a 120-s observe/record interval followed by a 30-s break interval to allow for placement to observe the next target participant. This sequence was repeated for the entire 15-min observation period with observations conducted for both children with ASD and NTD.

General study conditions. The *Buddy Game* was conducted daily during the second 15-min session of each classroom's individual 45-min outdoor recess period. Generalization sessions occurred during a 15-min free play/center session in each classroom at a time after the outside recess period. Maintenance data were collected 1-week after the termination of the *Buddy Game* intervention for each classroom. The specifics of each study phase have been described previously (cf. Morrier & Ziegler, 2018) and are briefly described below.

Pre-baseline. One month prior to the start of the study, all teachers were instructed to not plan for or conduct any organized activities, games, or curriculum that they may have been previously implementing during their outside recess time. Classrooms were also asked to stop singing the movement songs that were part of the *Buddy Game* to minimize the child participants' habitation to the independent variable, while increasing their motivation.

Baseline condition. During baseline, one lead teacher and two to four assistant teachers were present. Children were free to engage in any outdoor activity available to them. Each teacher was assigned to facilitate one zone on the playground, and to contact children in their zone according to regular program procedures. Unengaged children were matter-of-factly redirected to an ongoing playground activity. During baseline, teachers were instructed not to prompt, to teach, or to reinforce any child–child interactions, except those required to settle a sharing dispute over materials or physical space.

Buddy Game *condition.* Before each class' outside time, teachers informed all child participants who would be playing the *Buddy Game* when they went outside, but to play, they first needed to have buddies. Names were randomly chosen from a shaken bucket containing the names of children with ASD and an identical bucket containing names of NTD peers. The buddy pairs were randomly selected, announced, and visually placed together using each child's nametag on the "Buddy Board." The Buddy Board was also taken outside with the class as a visual reminder to the children of who was their buddy.

During the *Buddy Game* time, one lead teacher and multiple assistant teachers were present. The children were informed that, "It's time to play the *Buddy Game*. Find your buddy and come to the field together." When all participating children had gathered "in the field," the lead teacher informed them that they were going to play the *Buddy Game*. The intervention consisted of child dyads singing songs with gestures and movements, in front of one another, in a larger group setting. Songs were randomly chosen by children from a selection of cards with the song's title and related pictorial depiction. The lead teacher randomly chose a different child to select each song. After the children gathered in the field, they were given the instructions:

> Friends, we are going to sing some songs with our buddies. They are here on these cards. (The lead teacher randomly chose a child to pick a card; the child picked a card). Oh, *X* chose *song Y.* Ok, we are going to stand in front of our buddies to sing. Remember, we are dancing and singing with our buddies.

Teachers supported any child who needed assistance to stand in front of their buddy, forming two lines, using graduated guidance procedures (Wolery et al., 1992), by gently leading them by the hands to guide them into position, to begin the intervention. All teachers then stood behind the children to minimize the child participants' attention on them. If dyads drifted out of the two formed lines, or the lines drifted into a circle, children were not prompted to return to the original formation. Following this procedure, all seven songs were song. If time remained in the 15-min period, after the seventh song ended, the procedure was repeated. Teachers were instructed not to prompt, to teach, or to reinforce any child–child interactions that were not related to the intervention.

Generalization condition. In a 15-min indoor free play/center session following the *Buddy Game*, teacher behavior mirrored that of the baseline condition. During this condition, children were free to engage in any activity or with any toys that were available to them. Teachers were instructed to contact children using regular classroom procedures. Teachers were instructed to not prompt or teach to, or reinforce any child–child interactions, except those required to settle a sharing dispute over materials or physical space.

Maintenance condition. Approximately 1 week after the *Buddy Game* condition ended, follow-up data were collected to determine the maintenance of skills across time. Maintenance probes were conducted both during an outside recess period and the after recess free play/center period for each class. Settings and procedures mirrored both the baseline and generalization conditions.

Response Definition and Reliability

The dependent variable in this study was *discrete social bids from target participants with ASD to peers with NTD*. This was defined as an overt, observable verbal or non-verbal social communication bid from a child with ASD to a NTD peer where the target participant was facially oriented toward their peer and either vocalized and/or gestured to the peer. Examples of discrete social bids included (a) child saying name of peer; (b) child waving at another and simultaneously commenting on their shirt; and (c) child asking peer to play chase. The target participant with ASD was defined as the child whose behavior was being observed during the 120-s interval; while peers were defined as any other child who was not the target participant. Discrete social bids from children with ASD to NTD peers were used as the deciding behavior when making analytical decisions regarding exposure of each classroom to their respective intervention phase. This behavior was chosen as it was the most active and clinically significant target behavior for the participants with ASD. Significant increases in this behavior would indicate developmental changes in social communication and social interactions, the hallmark deficit that children with ASD experience (APA, 2013). Although not reported here, data were collected on the discrete social bids of NTD peers to children with ASD to further examine whether there was a clinical effect of the intervention on their social experiences with children with ASD (Sabey et al., 2020).

Interobserver agreement. Prior to the study's commencement, four staff with experience in coding data, including the first and second authors, independently scored videos of PS-aged children not affiliated with this study, at play in a free-play setting using the defined dependent variables. Data collection for this study began when each investigator achieved an agreement of 80% or greater with the first author on three consecutive video segments.

Across all study conditions, two of the four trained investigators simultaneously and independently observed and coded 38% of sessions to determine the consistency of measurement of the recorded behaviors between the observers. IOA was calculated using a mean count-per-interval IOA formula (Cooper et al., 2020, p. 115; Gast & Ledford, 2014, p. 146). Specifically, agreement between the total frequencies of responses recorded by two independent observers was calculated by dividing the total number of agreements between each observer's counts within each interval, by the total number of intervals scored. Observers were not blind to the study condition in which data were collected.

Overall study IOA was 89% (range: 80%–100%) between all observers and across all study phases. The average IOA in baseline was 88% (range: 80%–98%), 93.8% in intervention (range: 83%–100%), 89% in generalization in free play (range: 83%–95%), and 89% in maintenance (range: 84%–94%).

Data analysis. To determine the effect of the *Buddy Game*, frequency data were aggregated, or summed, across participants within each observation to see the effects of the *Buddy Game* within and across classroom cohorts. Previous research (Morrier & Ziegler, 2018) demonstrated that overall changes in social behavior were significantly influenced by ASD status and age was a non-significant factor. Visual analysis was used to examine the data with respect to level, trend, and variability (Gast & Ledford, 2014). To determine effect size, Tau-*U* (Parker et al., 2011) was calculated using a free online Tau-*U* calculator (Vannest et al., 2016). A Tau-*U* score of .20 can be considered a small change, .20 to .60 a moderate change, .60 to .80 a large change, and above .80 large to very large change (Vannest & Ninci, 2015). Tau-*U*

gain scores were also calculated to determine the magnitude of effect in each classroom.

Procedural fidelity

Adult interventionists' procedural fidelity. Two treatment fidelity checklists were developed for the treatment condition: one for the lead teacher and one for assistant teachers. These checklists were task analyses of the duties required for each teacher role during the intervention previously described. Fidelity of intervention implementation was assessed for each interventionist depending on their role, every second session in which they participated, using the same checklists across all study phases.

To obtain procedural fidelity, the interventionist had to pass two consecutive checklists with 100% fidelity. If an interventionist did not perform one duty on the checklist to fidelity, the checklist was considered failed. The observers were not blind to the study condition in which data were collected. Procedural fidelity was measured 50% of sessions during the study's baseline and maintenance phases in all three classrooms. Procedural fidelity during the intervention phase was measured during 40% of sessions in PK, 25% of sessions in EPS, and 33% of sessions in PS. Differences in percentages reflect a differing number of sessions needed to achieve fidelity by all interventionists in each classroom. During baseline sessions, no teachers demonstrated procedural fidelity. During intervention, eight lead teachers demonstrated 94% procedural fidelity over all assessed sessions. Seven lead teachers passed the procedural fidelity requirements by completing two intervention sessions to 100% fidelity. One lead teacher failed to comply with procedural fidelity requirements by failing the procedural fidelity of one of the two intervention sessions in which they participated.

The six participating assistant teachers demonstrated 94% procedural fidelity over all assessed sessions. Four assistant teachers passed the procedural fidelity requirements by completing two intervention sessions to 100% fidelity. One assistant teacher passed the procedural fidelity requirements by completing the one intervention session they participated in to 100% fidelity. One assistant teacher passed the procedural fidelity requirements by completing two intervention sessions to 100% fidelity, and failing one, resulting in 67% fidelity in all completed sessions. Procedural fidelity checklists are available from the first author upon request.

Results

Discrete Social Bids From Children With ASD to Peers With NTD

Results for discrete social bids from children with ASD to NTD peers are presented in Figure 1. Overall, the results show an increase in discrete social bids from children with ASD in the all three classrooms following the introduction of the *Buddy Game* intervention. Tau-*U*s indicate a moderate effect size overall (i.e., .58). After the implementation of the *Buddy Game*, participants with ASD in all three classrooms increased their frequency of discrete social bids toward NTD peers from an average of 1.84 at baseline to an average of 5.11 during the *Buddy Game*.

PK demonstrated an increased from an average of .20 discrete social bids in five sessions (range: 0–1) in baseline to an average of 5.50 discrete social bids in four sessions during intervention (range: 1–10). PK classroom demonstrated an immediacy of intervention effect with a change in level and an upward trend while ending on a slight downward trend over these four sessions. EPS demonstrated variability in discrete social bids before increasing from an average of 1.78 discrete social bids in baseline (range: 0–5) to an average of 5.13 during intervention (range: 0–12). EPS classroom also demonstrated an immediacy of intervention effect, a change in level and an upward trend through an increase of discrete social bids. PS demonstrated large variability throughout baseline but showed a downward trend over the last three sessions with an average of 2.50 discrete social bids (range: 0–9), before moving into intervention, with an average of 4.83 (range: 0–10). PS classroom demonstrated an upward trend in discrete social bids over the six intervention sessions. The break in the intervention phase seen in Tiers 2 and 3 was due to a 3-day weekend, resulting in three consecutive days that the intervention was not implemented. Tau-U effect sizes ranged from moderate for EPS and PS classrooms to very large for PK classroom (i.e., .95 for PK, .54 for EPS, and .38 for PS).

Generalization of discrete social bids from children with ASD to their NTD peers was mixed across classrooms, demonstrating an average of 2.00 discrete social bids across all three classrooms. Participants with ASD in PK demonstrated discrete social bids to NTD peers during center time generalization sessions ($M = 5.75$; range: 0–8). Children with ASD in PS demonstrated some generalized discrete social bid behaviors ($M = 3.80$; range: 0–8) during free play. Children with ASD in EPS classrooms did not generalize ($M = .13$; range: 0–1) discrete social bid behaviors. Maintenance of this behavior was also mixed, with only the children with ASD in PK classroom demonstrating discrete social bid behaviors above baseline levels, both during outdoor recess and free play in different settings and at different times.

Social Validity Reported by Participating Children

Of the 28 child participants who completed the study (ASD = 9, NTD = 19), five children with ASD indicated that they *really liked* the *Buddy Game*, four children with ASD indicated that they *liked* the *Buddy Game*, and no children

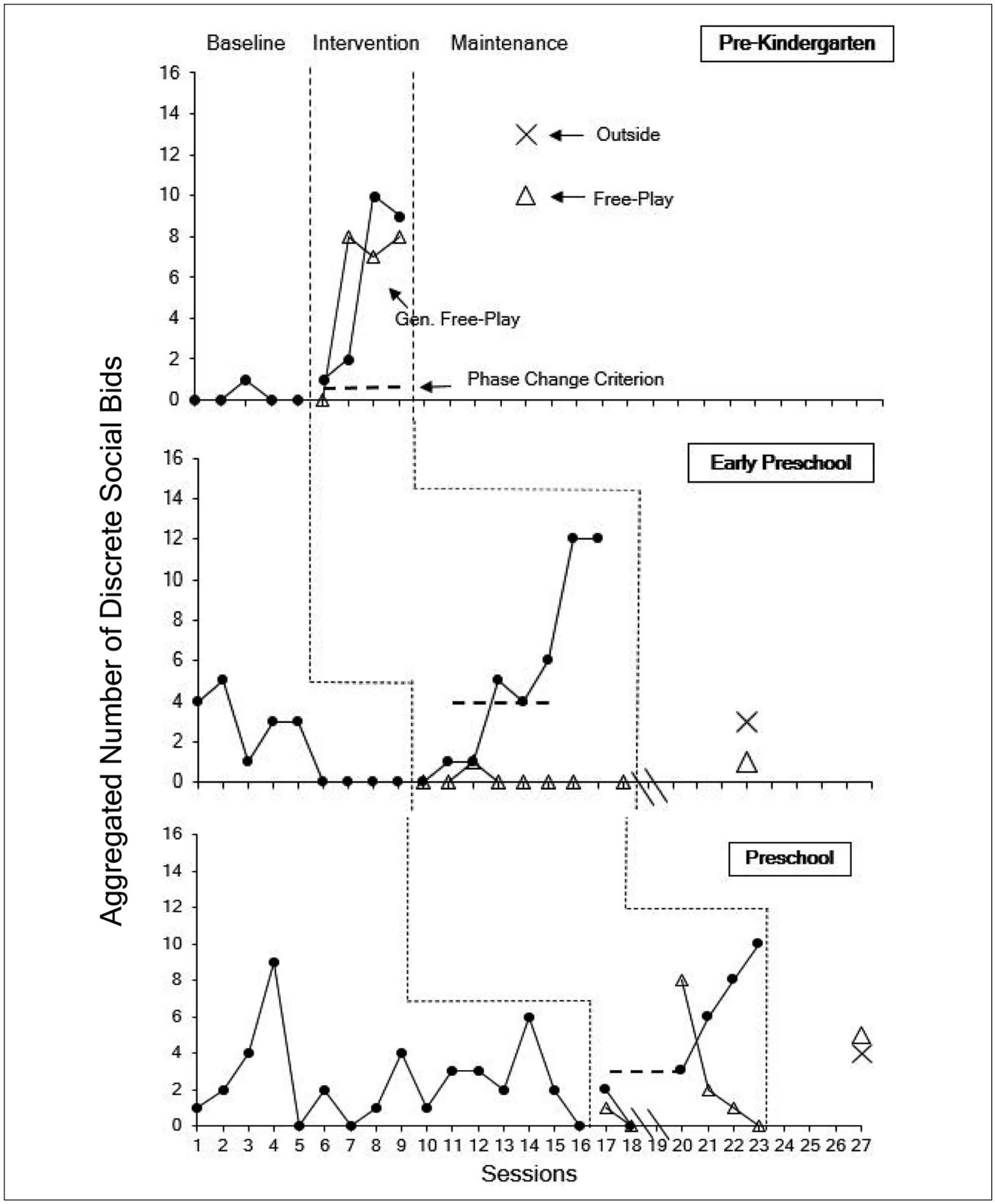


Figure 1. Discrete social bids from children with ASD to their peers with NTD over three classrooms in baseline, intervention, generalization, and maintenance.

Note. The break seen in EPS and PS was due to a 3-day weekend. ASD = autism spectrum disorder; NTD = neurotypical development; EPS = early preschool; PS = preschool.

with ASD indicated that they *did not like* the *Buddy Game*. Fourteen NTD children indicated that they *really liked* the *Buddy Game*, four indicated that they *liked* the *Buddy Game*, and one child indicated that they *did not like* the *Buddy Game*, because they found it "boring."

Discussion

A multiple baseline deign across PS classrooms was used to investigate the effects of the *Buddy Game* on the discrete social bid behavior between children with ASD and their NTD peers. Results demonstrated that the *Buddy Game* increased the number of discrete social bids by children with ASD across three PS classrooms; however, due to high variability in PS classroom's data before the intervention was introduced, a functional relation over all three classrooms was not demonstrated. This variability was likely due to the aggregation of the number of social bids, with the culmination of low levels of social bids from all the participants presented as a change in magnitude.

This study confirms findings of earlier studies (Boyd et al., 2008; McGee et al., 1999, 2001; Morrier et al., 2009; Stahmer et al., 2011; Strain & Bovey, 2011), demonstrating that interventions for children with ASD can be effectively embedded into naturally occurring play activities. The outcomes of this current study indicate when an intervention that focuses on natural modeling and imitation is implemented as a structured play curriculum, discrete social bids from children with ASD to peers with NTD increase.

In the current investigation, child participants randomly choose the movement songs they sung providing an element of child choice often found in NDBI interventions (Bruinsma et al., 2020; Schreibman et al., 2015). The results support previous findings that children with ASD can increase language and social skills when intervention is provided during games that included child choice (Carter, 2001; Miller & Almon, 2009; Morrier et al., 2009; Morrier & Ziegler, 2018). This study expands upon previous social interaction research between children with ASD and their peers with NTD, which primarily focused on older PS- or elementary-aged students, often only targeting two to three children at a time (Boyd et al., 2008; Harper et al., 2008; McEvoy et al., 1988; Owen-DeSchryver et al., 2008), by studying a total of 32 children (ASD = 9, NTD = 23) across three different classrooms in an inclusive PS setting. Furthermore, this study expands upon earlier research (McGee et al., 1992), to include children both younger and older children, aged 2 years 6 months to 6 years 6 months. This study moves the research focus from the implementation of an intervention with individual participants to implementation in a neurodiverse group setting.

When reviewing the research base, in previous studies that paired children with ASD and NTD (Harper et al., 2008; Koegel et al., 2005, 2006; Laushey & Heflin, 2000), participants were specifically trained how to interact with their peers with different neurodevelopment. McGee et al. (1992) highlighted a weakness in the concept of training children to teach peers when they described a decrease in fidelity between PS peer dyads where participants stopped demonstrating the intervention behavior. The current study demonstrates that modeling and imitation that occurs naturally within play can positively affect the social communication and interactions of children with ASD without the need for specific training, which can inadvertently set peers apart.

During social interactions between peers with differing neurodevelopmental patterns various motivating operations, as well as behaviors that have evocative or abative effects may be occurring. Children with NTD who are taught to prompt peers with ASD in a certain manner may find this unnatural social bid aversive (e.g., that they are "directing" their peer in a manner that they would not do naturally). Their peers with ASD may not socially reinforce these efforts by not responding, which may put these behaviors on extinction (i.e., receive no reinforcement), or even punish them resulting in NTD social bids stopping, because the lack of response from the child with ASD may be perceived as intentional or "being mean." Through less engagement of their NTD peer, a child with ASD may lose opportunities to have their own behavior socially or tangibly reinforced. This study's outcomes highlight valuable benefits that children with ASD gain from a motivating and reinforcing integrated learning environment with ample opportunities to spontaneously observe, imitate, and practice essential social communication skills.

This study failed to demonstrate the generalization of the behavior change obtained during intervention to the free play settings in the 2- and 3-year-old classrooms. The demonstration of minimal maintenance in these classrooms confirms previous findings that a lack of structured activity time lends children with ASD to experience difficulties engaging in play, suggesting that engagement in non-socially interactive behaviors may be more motivating than peer interactions (Peeters, 1997; Veale, 1998) in younger children.

It should be noted that PK's center time differed from that of free play in EPS and PS. Although participants in EPS and PS had access to the same materials in their free play settings, their access to these materials was unstructured and unlimited, providing less support or opportunities through necessity for discrete social bids from children with ASD to peers with NTD in those classrooms.

One week after the termination of intervention, outside maintenance probe data for each classroom were collected. During the PK's maintenance data collection, teachers encouraged the children to participate in a game of freeze tag. Although this game did not involve movement songs, nor focused on sustained natural modeling and imitation, it

was structured. Children in EPS and PS were not directed to play in a structured manner during their classrooms' outside maintenance data collection. These findings underline the importance of sustained structured play for the social communication development of children with ASD. Due to PK participants' graduation from the center shortly after the first maintenance probes were taken, further maintenance data could not be collected.

Limitations and Future Directions

Although the current investigation resulted in positive results for the directly targeted social behavior of children with ASD (i.e., number of discrete social bids to NTD peers), limitations may result in future meaningful research questions. First, the time interval of 2 min allotted to observe each target participant during data collection across all phases was a limitation. Although this observation interval ensured that all target participants were equally observed, supporting the reliability of the behavioral data, such a limited time interval may have restricted the true representation of a target participant's behavior over the full 15-min observation period. A child who engaged in solitary play during their 120-s observation interval may have engaged a peer who walked by them at the end of that interval and remained in reciprocal play with that peer for the duration of the observation period. This, however, would not reflect in the data. A point for future consideration is the use of the formula ASD Gives + Receipts from NTD during an observation period, or the sum of discrete social bids from a child with ASD and the receipts of discrete social bids from their peers. This may provide a more valid presentation of the social communication behavior and experiences of children with ASD.

Second, the study's failure to demonstrate both the generalization and maintenance of the targeted social behavior over all three classrooms begs further investigation. Previous research on the *Buddy Game* indicated that having an ASD diagnosis was a significant factor for increasing social behavior across study phases (Morrier & Ziegler, 2018), although age of child did not make a significant contribution to the observed increases. Further maintenance measures should be taken to demonstrate the maintenance of the invention's effect over time. The graduation of PK made this impossible in this study. Furthermore, as both generalization and maintenance were demonstrated in the structured environments of PK's center time and recess settings, future research may evaluate the effectiveness of the intervention specifically in structured settings or compare its effectiveness in structured and non-structured environments.

Future research may also evaluate age, or time spent in treatment as determining factors in the effectiveness of the intervention. The aggregation of participants' discrete social bids to their NTD peers is a significant limitation. Some participants may have responded to the intervention differently than others and could further inform the research and intervention development. The study's observers were not blinded to the study. Data and collection experiences were never discussed between team members; however, this could not guarantee a total lack of measurement bias.

Finally, the exclusion of a teacher social validity measure is a limitation. Regardless of defined role and daily duties, information gathered from an intervention implementer is valuable in guiding further research and innovation.

Implications for Practitioners

Structured, cooperative play with a focus on natural modeling and imitation appears to be an effective means of increasing social communication of children with ASD of varying ages. Limits that structured play provide may support children with ASD to focus on their partners in play and the social information that these partners convey through their interactions. These findings have significant implications for early intervention practices, parent training, and educational curricula. Incorporating structured, cooperative play that includes peers with NTD into early intervention may greatly enhance the efficacy of current practices on the social communication development of children with ASD.

Author's Note

Sonja M. T. Ziegler is now at the Department of Child and Adolescent Psychiatry of the Region of Southern Denmark. This article uses a subset of data from a larger data set that was published in the work of Morrier and Ziegler (2018). The findings and conclusions in this study are those of the authors and do not necessarily present the official position of Emory University School of Medicine or Georgia State University.

Acknowledgments

The authors would like to thank the children, families, and staff from the Early Emory Center for Child Development and Enrichment who participated in this study. They extend their special thanks to Brittni Williams, MEd, BCBA, and Devika Persaud, MEd, BCBA, for collecting data with them.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

ORCID iDs

Sonja M. T. Ziegler https://orcid.org/0000-0003-2113-2080

Michael J. Morrier https://orcid.org/0000-0002-9219-437X

References

American Psychiatric Association. (2013). *Diagnostic and statistical manual for mental disorders* (5th ed.). https://doi.org/10.1176/appi.books.9780890425596

Anderson, A., Moore, D. W., Godfrey, R., & Fletcher-Flinn, C. M. (2004). Social skills assessment of children with autism in free-play situations. *Autism*, *8*(4), 369–385. https://doi.org/10.1177/1362361304045216

Baer, D. M., Wolf, M. M., & Risley, T. R. (1968). Some current dimensions of applied behaviour analysis. *Journal of Applied Behavior Analysis*, *1*(1), 91–97. https://doi.org/10.1901/jaba.1968.1-91

Boyd, C. M., Fraiman, J. L., Hawkins, K. A., Labin, J. M., Sutter, M. B., & Wahl, M. R. (2008). Effects of the STAR intervention program on the interactions between campers with and without disabilities during inclusive summer day camp activities. *Education and Training in Developmental Disabilities*, *43*(1), 91–101.

Bruinsma, Y., Minjarez, M., Schreibmam, L., & Stahmer, A. (2020). *Naturalistic developmental behavior interventions for autism spectrum disorder*. Paul H. Brookes.

Carter, C. M. (2001). Using choice with game play to increase language skills and interactive behaviors in children with autism. *Journal of Positive Behavior Interventions*, *3*(3), 131–151. https://doi.org/10.1177/109830070100300302

Cooper, J. O., Heron, T. E., & Heward, W. L. (2020). Improving and assessing the quality of behavioral measurement. In J. O. Cooper, T. E. Heron, & W. L. Heward (Eds.), *Applied Behavior Analysis* (3rd ed., pp. 101–121). Pearson Education.

Council on School Health. (2013). The crucial role of recess in school. *Pediatrics*, *131*(1), 183–188. https://doi.org/10.1542/peds.2012-2993

Gast, D. L., & Ledford, J. R. (Eds.). (2014). *Single case research methodology: Applications in special education and behavioral sciences* (2nd ed.). Routledge.

Georgia Department of Early Care and Learning. (2016). *Rules for childcare learning centers* (Chapter 591-1-1). http://decal.ga.gov/documents/attachments/CCLCRulesandRegulations.pdf

Harper, C. B., Symon, J. B. G., & Frea, W. D. (2008). Recess is time-in: Using peers to improve social skills of children with autism. *Journal of Autism and Developmental Disorders*, *38*(5), 815–826. https://doi.org/1007/s10803-007-0449-2

Ingram, D. H., Mayes, S. D., Troxell, L. B., & Calhoun, S. L. (2007). Assessing children with autism, mental retardation, and typical development using the Playground Observation checklist. *Autism*, *11*(4), 311–319. https://doi.org/10.1177%2F1362361307078129

Kasari, C., Rotheram-Fuller, E., Locke, J., & Gulsrud, A. (2012). Making the connection: Randomized controlled trial of social skills at school for children with autism spectrum disorders. *Journal of Child Psychology and Psychiatry*, *53*(4), 431–439. https://doi.org/10.1111/j.1469-7610.2011.02493.x

Koegel, R. L., Openden, D., Fredeen, R., & Koegel, L. K. (2006). The basics of pivotal response training treatment. In R. L. Koegel & L. K. Koegel (Eds.), *Pivotal response treatments for autism: Communication, social and academic development* (pp. 3–30). Paul H Brookes.

Koegel, R. L., Werner, G. A., Vismara, L. A., & Koegel, L. K. (2005). The effectiveness of contextually supported play date interactions between children with autism and typically developing peers. *Research and Practice for Persons With Severe Disabilities*, *30*(2), 93–102. https://doi.org/10.2511/rpsd.30.2.93

Kohler, F. W., Greteman, C., Raschke, D., & Highnam, C. (2007). Using a buddy skills package to increase the social interactions between a preschooler with autism and their peers. *Topics in Early Childhood Special Education*, *27*(3), 155–163. https://doi.org/10.1177/02711214070270030601

Lang, R., Kuriakose, S., Lyons, G., Mulloy, A., Boutot, A., Britt, C., Caruthers, S., Ortega, L., O'Reilly, M., & Lancioni, G. (2011). Use of school recess time in the education and treatment of children with autism spectrum disorders: A systematic review. *Research in Autism Spectrum Disorders*, *5*(4), 1296–1305. https://doi.org/10.1016/j.rasd.2011.02.012

Laushey, K. M., & Heflin, L. J. (2000). Enhancing social skills of kindergarten children with autism through the training of multiple peers as tutors. *Journal of Autism and Developmental Disorders*, *30*(3), 183–193. https://doi.org/10.1023/A:1005558101038

Lord, C., Luyster, R. J., Gotham, K., & Guthrie, W. (2012). *Autism Diagnostic Observation Schedule-2nd edition (ADOS-2) manual (Part II): Toddler module*. Western Psychological Services.

Lord, C., Rutter, M., DiLavore, P. C., Risi, S., Gotham, K., & Bishop, S. L. (2012). *Autism Diagnostic Observation Schedule-2nd edition (ADOS-2) manual (Part 1): Modules 1-4*. Western Psychological Services.

McEvoy, M. A., Nordquist, V. M., Twardosz, S., Heckman, K. A., Wheby, J. H., & Denny, R. K. (1988). Promoting autistic children's peer interaction in an integrated early childhood setting using affection activities. *Journal of Applied Behavior Analysis*, *21*(2), 193–200. https://doi.org/10.1901/jaba.1988.21-193

McGee, G. G., Almeida, M. C., Sulzer-Azaroff, B., & Feldman, R. S. (1992). Promoting reciprocal interactions via peer incidental teaching. *Journal of Applied Behavior Analysis*, *25*(1), 117–126. https://doi.org/10.1901/jaba.1992.25-117

McGee, G. G., & Daly, T. (1999). Prevention of problem behaviors in preschool children. In A. C. Repp & R. H. Horner (Eds.), *Functional analysis of problem behaviors: From effective assessment to effective support* (pp. 171–196). Wadsworth.

McGee, G. G., Morrier, M. J., & Daly, T. (1999). An incidental teaching approach to early intervention for toddlers with autism. *Journal of the Association for Persons with Severe Handicaps*, *24*(3), 133–146. https://doi.org/10.2511/rspd.24.3.133

McGee, G. G., Morrier, M. J., & Daly, T. (2001). The Walden early childhood programs. In J. S. Handleman & S. L. Harris (Eds.), *Preschool education programs for children with autism* (2nd ed., pp. 157–190). PRO-ED.

McNamara, L., Lakman, Y., Spadafora, N., Lodewyk, K., & Walker, M. (2018). Recess and children with disabilities: A mixed-methods pilot study. *Disability and Health Journal*, *11*(4), 637–643. https://doi.org/10.1016/j.dhjo.2018.03.005

Miller, E., & Almon, J. (2009). *Crisis in the kindergarten: Why children need play in school*. Alliance for Childhood. http://www.allianceforchildhood.org

Morrier, M. J., McGee, G. G., & Daly, T. (2009). Effects of toy selection and arrangement on the social behaviors of an inclusive group of preschool-aged children with and without autism. *Early Childhood Services*, *3*(2), 157–177. https://doi.org/10.1007/s1080

Morrier, M. J., & Ziegler, S. M. T. (2018). I wanna play too: Factors related to changes in social behavior for children with and without autism spectrum disorder after implementation of a structured outdoor play curriculum. *Journal of Autism and Developmental Disorders*, *48*(7), 2530–2541. https://doi.org/10.1007/s10803-018-3523-z

Owen-DeSchryver, J. S., Carr, E. G., Cale, S. I., & Blakeley-Smith, A. (2008). Promoting social interactions between students with autism spectrum disorders and their peers in inclusive school settings. *Focus on Autism and Other Developmental Disabilities*, *23*(1), 15–28. https://doi.org/10.1177/1088357608314370

Parker, R. I., Vannest, K. J., Davis, J. L., & Sauber, S. B. (2011). Combining nonoverlap and trend for single-case research: Tau-U. *Behavior Therapy*, *42*(2), 284–299. https://doi.org/10.1016/j.beth.2010.08.006

Peeters, T. (1997). *Autism: From theoretical understanding to educational intervention*. Singular.

Rutter, M., Le Couteur, A., & Lord, C. (2003). *Autism diagnostic interview—Revised*. Western Psychological Services.

Sabey, C., Ross, S., & Goodman, J. (2020). Beyond topography: Addressing the functional impact of social skills training for students with autism. *Educational Psychology in Practice*, *36*(2), 133–148. https://doi.org/10.1080/02667363.2019.1703650

Schreibman, L., Dawson, G., Stahmer, A. C., Landa, R., Rogers, S. J., McGee, G. G., Kasari, C., Ingersoll, B., Kaiser, A. P., Bruinsma, Y., McNerney, E., Wetherby, A., & Hallady, A. (2015). Naturalistic developmental behavioral interventions: Empirically validated treatments for autism spectrum disorder. *Journal of Autism and Developmental Disorders*, *45*(8), 2411–2428. https://doi.org/10.1007/s10803-015-2407-8

Shaw, K. A., Maenner, M. J., Baio, J., Washington, A., Christensen, D. L., Wiggins, L. D., Pettygrove, S., Andrews, J. G., White, T., Rosenberg, C. R., Constantino, J. N., Fitzgerald, R. T., Zahorodny, W., Shenouda, J., Daniels, J. L., Salinas, A., Durkin, M. S., & Dietz, P. M. (2020). Early identification of autism spectrum disorder among children aged 4 years—Early autism and developmental disabilities monitoring network, six sites, United States, 2016. *Morbidity and Mortality Weekly Report, Surveillance Summaries*, *69*(3), 1–11. https://doi.org/10.15585%2Fmmwr.ss6903a1

Sparrow, S., Ciccetti, D., & Saulnier, C. (2016). *Vineland Adaptive Behavior Scales, manual* (3rd ed.). Pearson.

Stahmer, A. C., Akshoomoff, N., & Cunningham, A. B. (2011). Inclusion for toddlers with autism spectrum disorders: The first ten years of a community program. *Autism*, *15*(5), 625–641. http://doi.org/10.1177/1362361310392253

Strain, P. S., & Bovey, E. H., II. (2011). Randomized, controlled trial of the LEAP model of early intervention for young children with autism spectrum disorders. *Topics in Early Childhood Special Education*, *31*(3), 133–154. https://doi.org/10.1177%2F0271121411408740

Vannest, K. J., & Ninci, J. (2015). Evaluating intervention effects in single-case research designs. *Journal of Counseling & Development*, *93*(4), 403–411. https://doi.org/10.1002/jcad.12038

Vannest, K. J., Parker, R. I., Gonen, O., & Adiguzel, T. (2016). *Single case research: Web based calculators for SCR analysis* (Version 2.0) [Web-Based Application]. A&M University. http://www.singlecaseresearch.org/

Veale, T. (1998). *May I have your attention please? Language and learning lessons from one child with autism* [Unpublished doctoral dissertation]. University of Cincinnati.

Wolery, M., Ault, M. J., & Doyle, P. M. (1992). *Teaching students with moderate to severe disabilities: Use of response prompting strategies*. Longman.